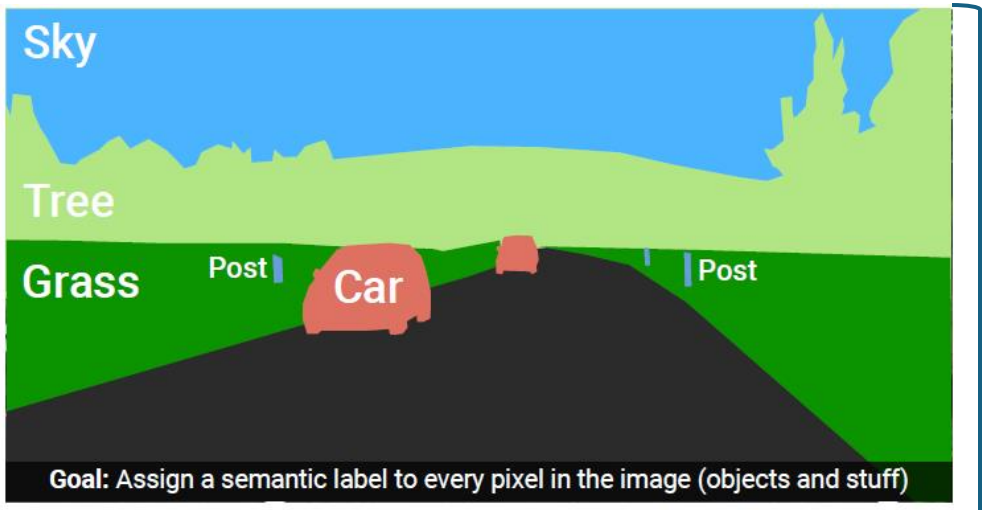
Segmentation and Detection

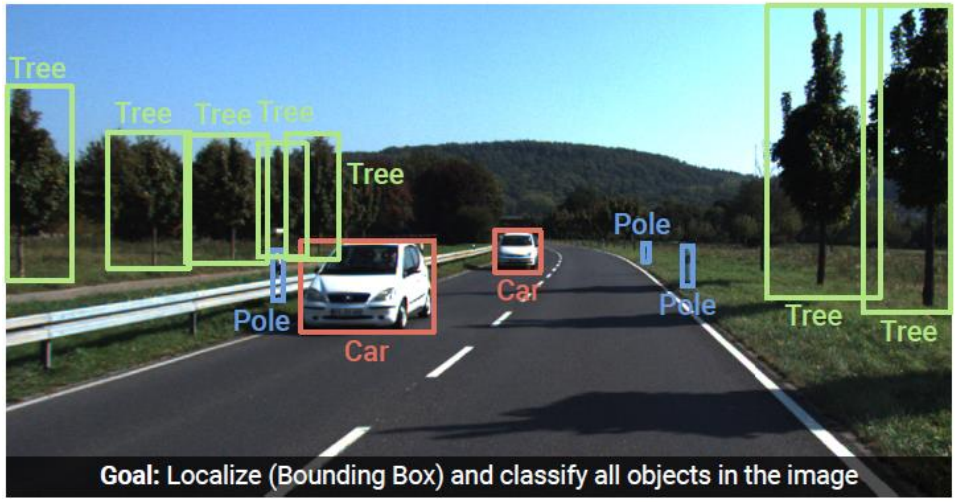
Computer Vision

Image Understanding (Recognition)

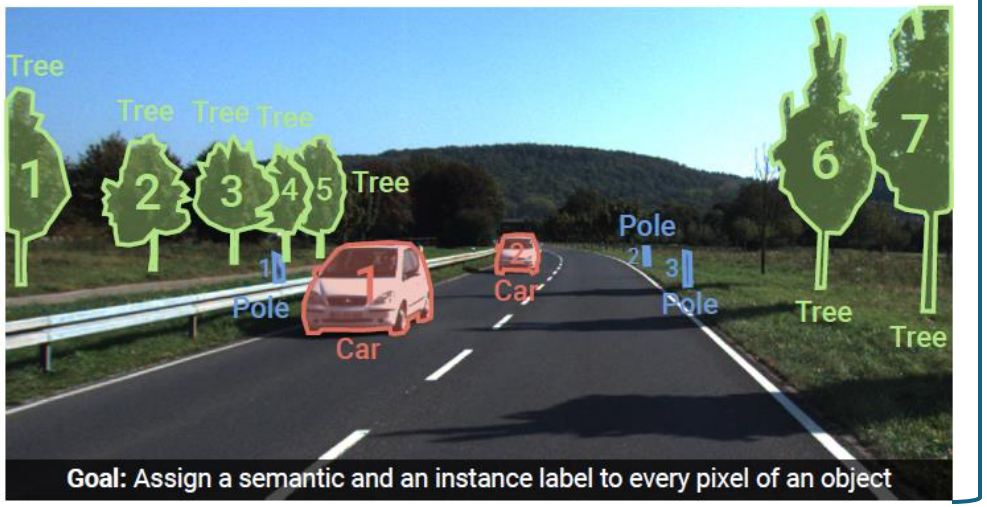
Image Classification



Semantic Segmentation



Object Detection



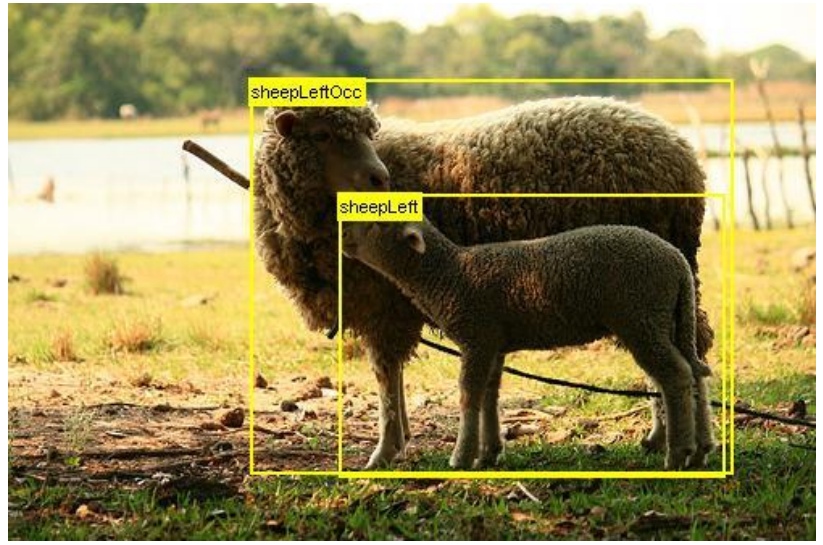
Instance Segmentation

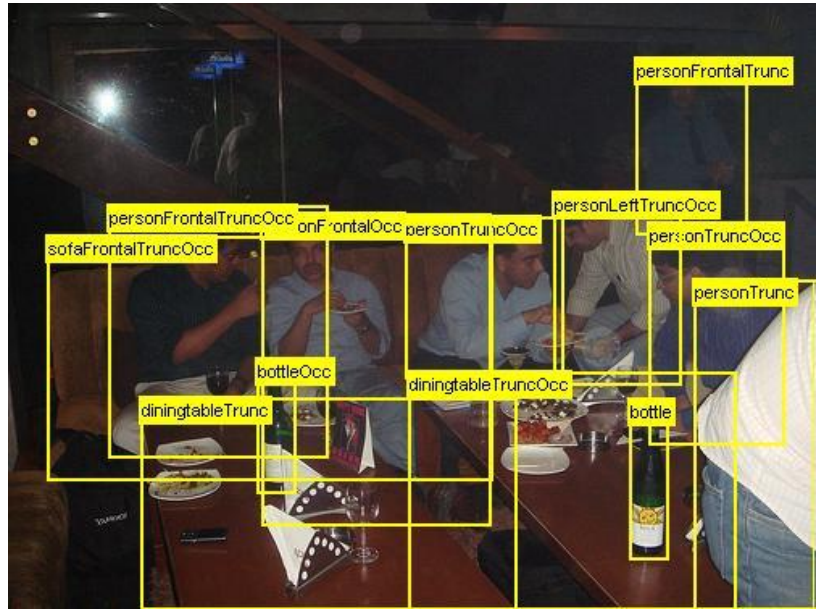
combination of both: panoptic segmentation

A Few More Image Data Sets

PASCAL VOC Data Set

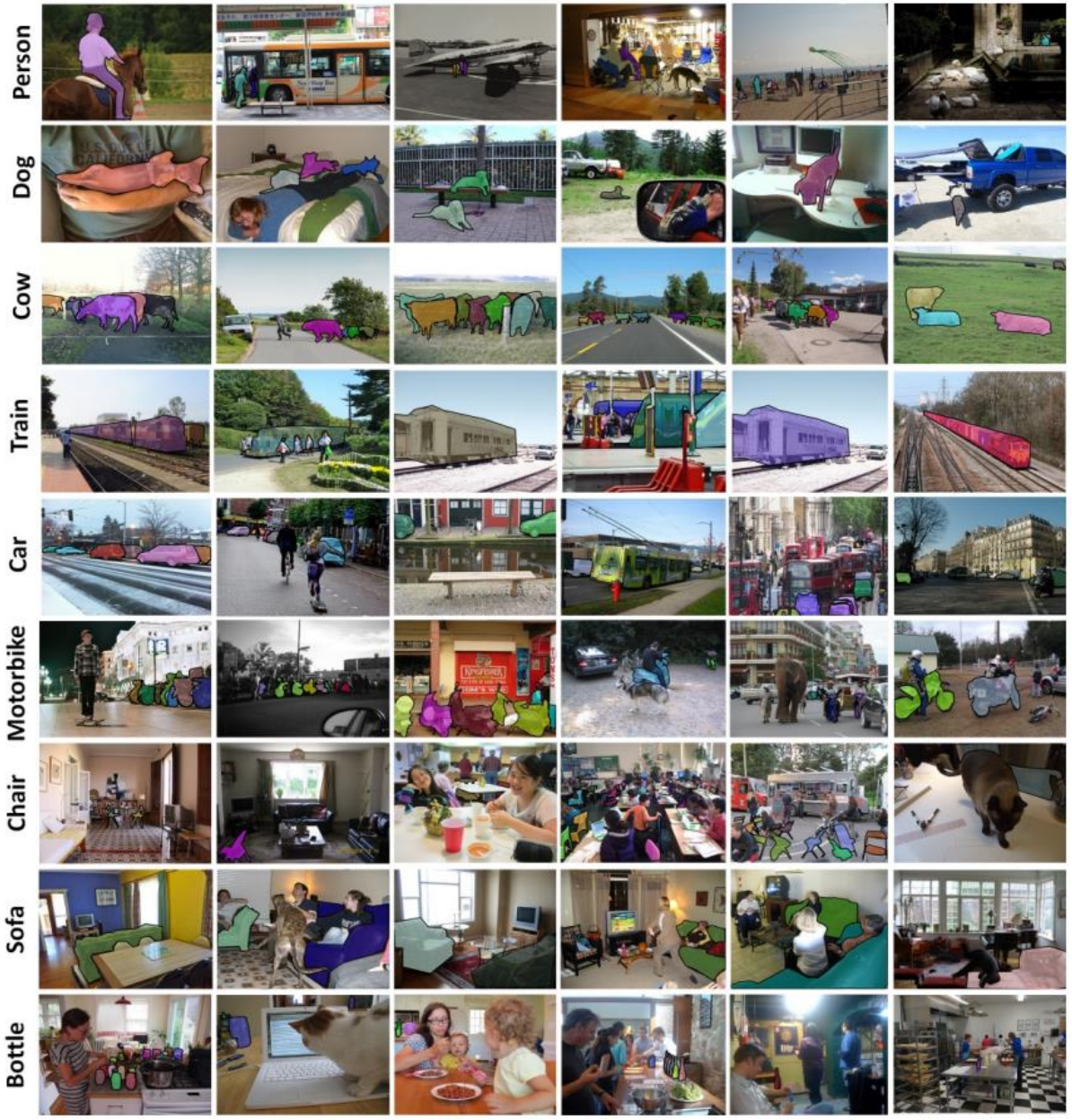
- PASCAL Visual Object Class challenge
- widely used as benchmark for object detection and semantic segmentation
- 20 object categories such as person, sofa, sheep, car, ...
- 11530 annotated images
- available annotations: pixel-level segmentation, bounding boxes, object classes





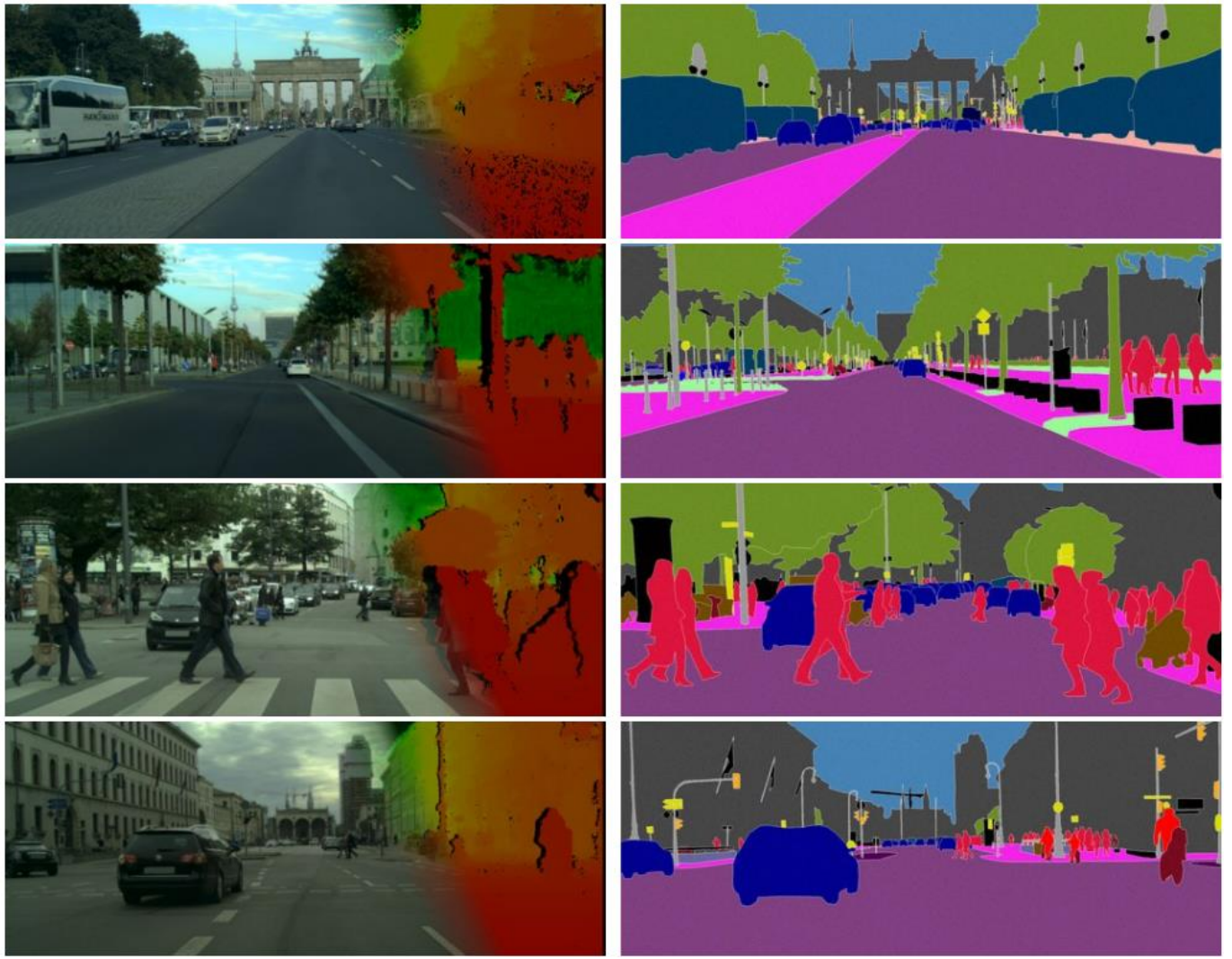
MS COCO Data Set

- Microsoft Common Objects in Context
- images of complex everyday scenes containing common objects in their natural context
- 91 objects types
- 2.5 million annotated instances in 328k images → instance segmentation



Cityscapes Data Set

- goal: semantic understanding of urban street scenes (captured in 50 cities)
- pixel annotations for 30 classes (person, car, building, …)
- 5000 fine-annotated and 20000 coarse-annotated images



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Semantic Segmentation

Object Segmentation from DINO

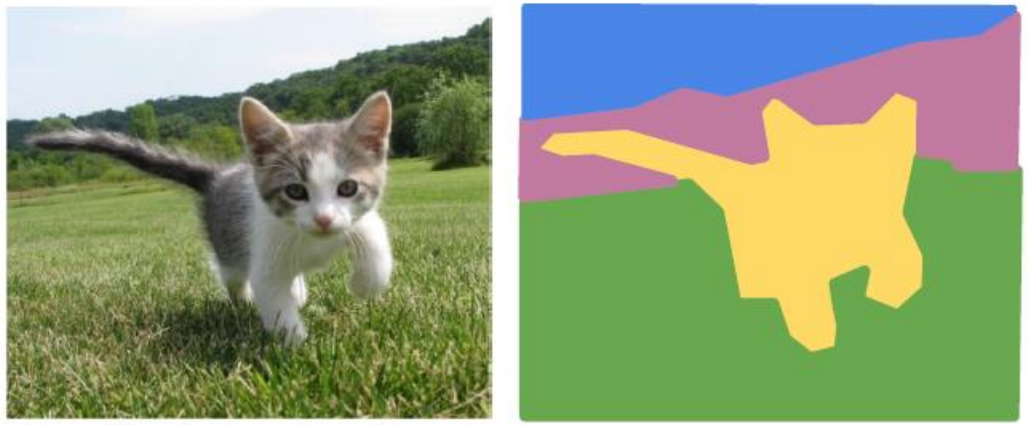
thresholding self-attention map of last layer:

source

not a full segmentation mask though …

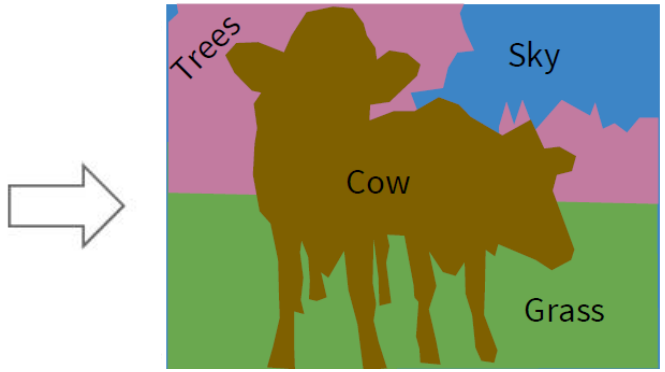
Classification of Each Pixel

segmentation: no objects, just pixels



GRASS, CAT, TREE,
SKY, ...

Paired training data: for each training image, each pixel is labeled with a semantic category.



At test time, classify each pixel of a new image.

minimize sum over classification losses
(cross-entropy loss at every output pixel)

Idea: Fully Convolutional, No Downsampling

Input:
3 x H x W

Conv → Conv → Conv → Conv → argmax

stride 1,
padding
→ small kernels

feature maps → Convolutions:
D x H x W

Scores:
C x H x W

categories

Predictions:
H x W

replace flattened, fully-connected classification layers with 1×1 convolutions
→ maintain spatial relationships and enable pixel-wise classification:
conversion of feature maps into classification heat maps (one for each class)

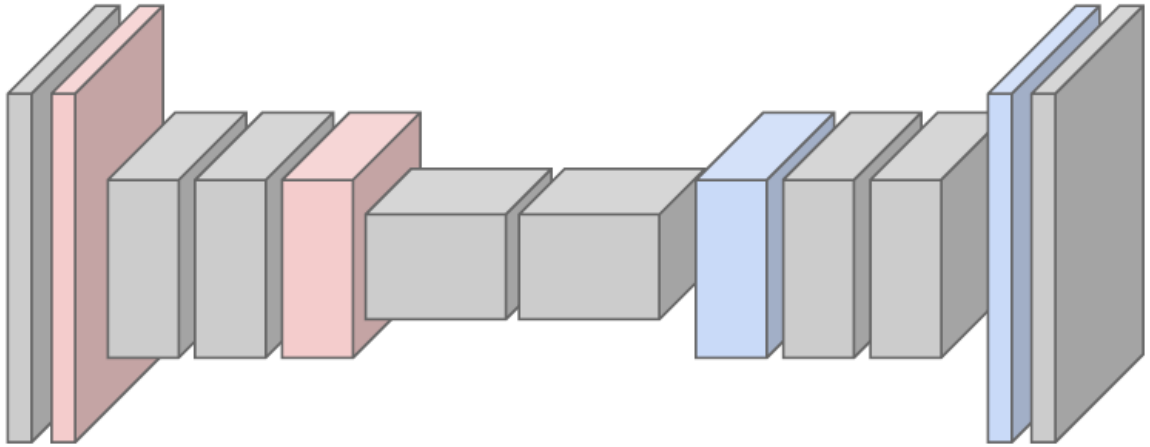
but no downsampling means small receptive field and no hierarchical learning

Upsampling to the Rescue

typical spatial feature extraction: convolution and pooling layers → downsampling

upsampling layers to restore original image size

Input:
3 x H x W



C x H x W

pixel-wise classification

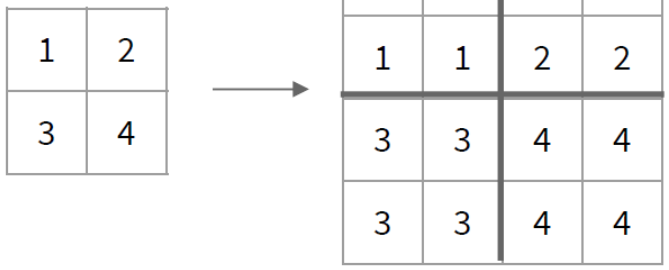
Predictions:
H x W

different options for down- (pooling, strided convolution) and upsampling ...

Reverse Pooling

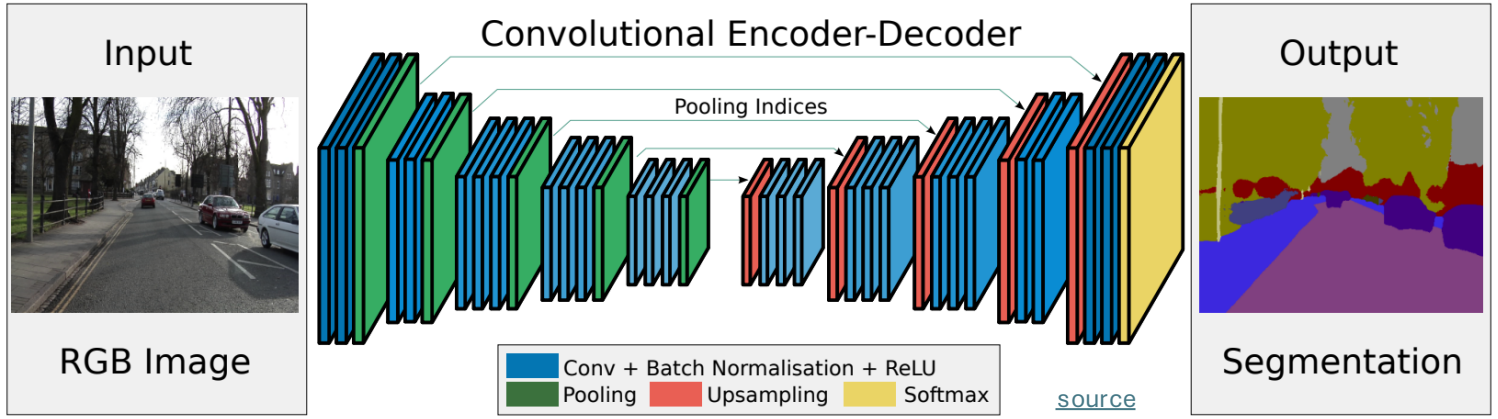
resampling (no learned parameters)

Nearest Neighbor

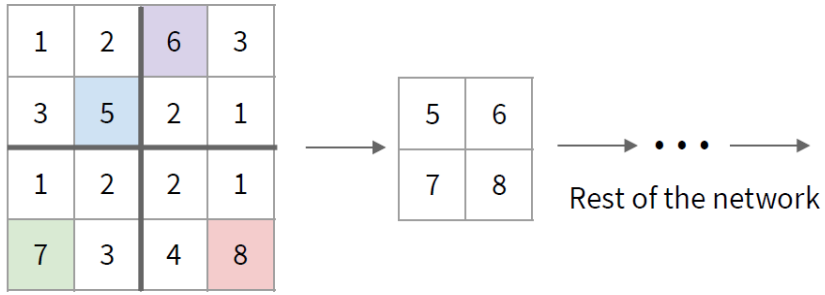


Input: 2 x 2 Output: 4 x 4

unpooling (recording max positions from pooling)

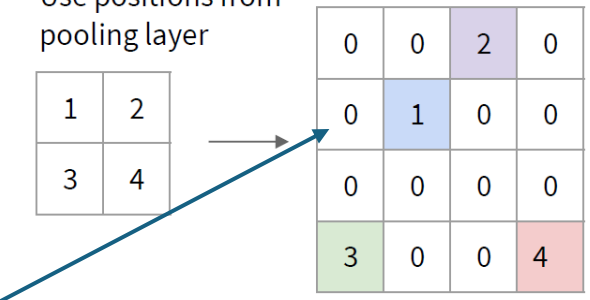


Max Pooling
Remember which element was max!



Input: 4 x 4 Output: 2 x 2

Max Unpooling
Use positions from pooling layer

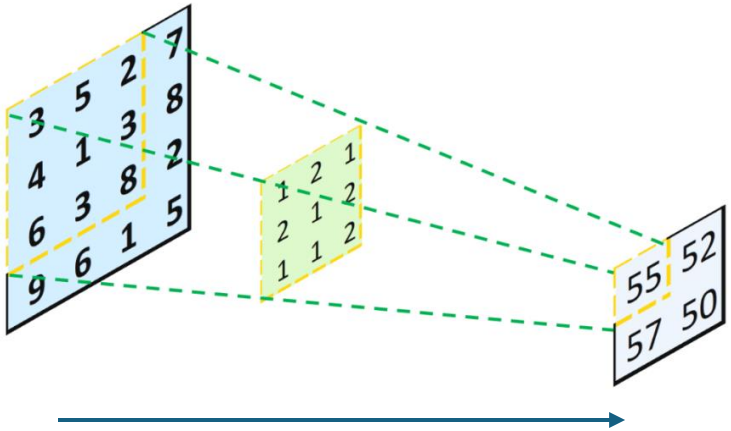


Input: 2 x 2 Output: 4 x 4

filled with learned parameters in subsequent convolution

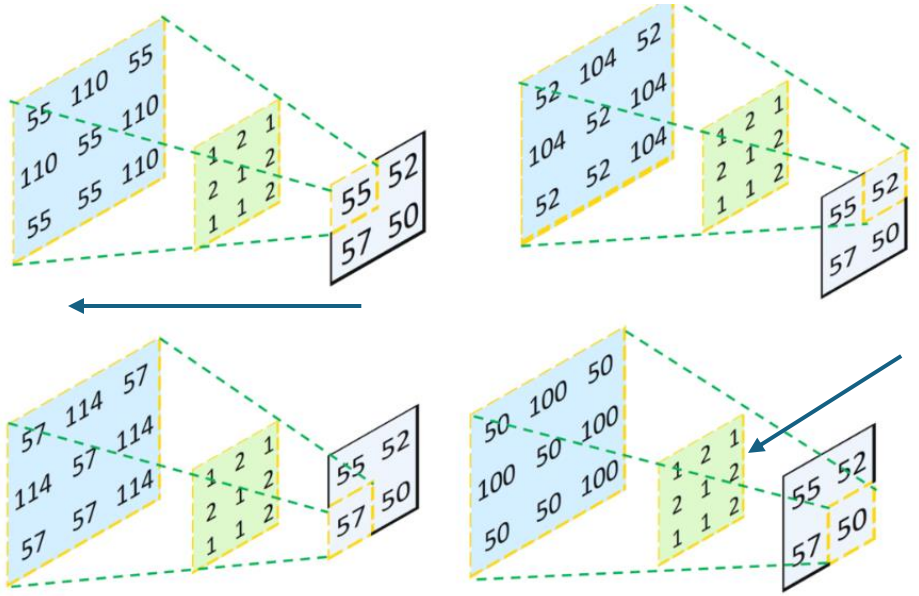
Reverse Convolution

convolution

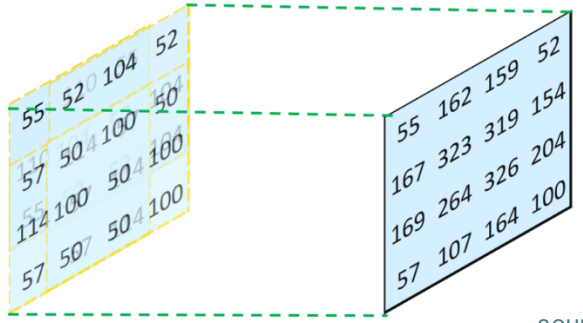


not inverse convolution
→ additional learned parameters

transposed convolution



combine and sum overlaps:



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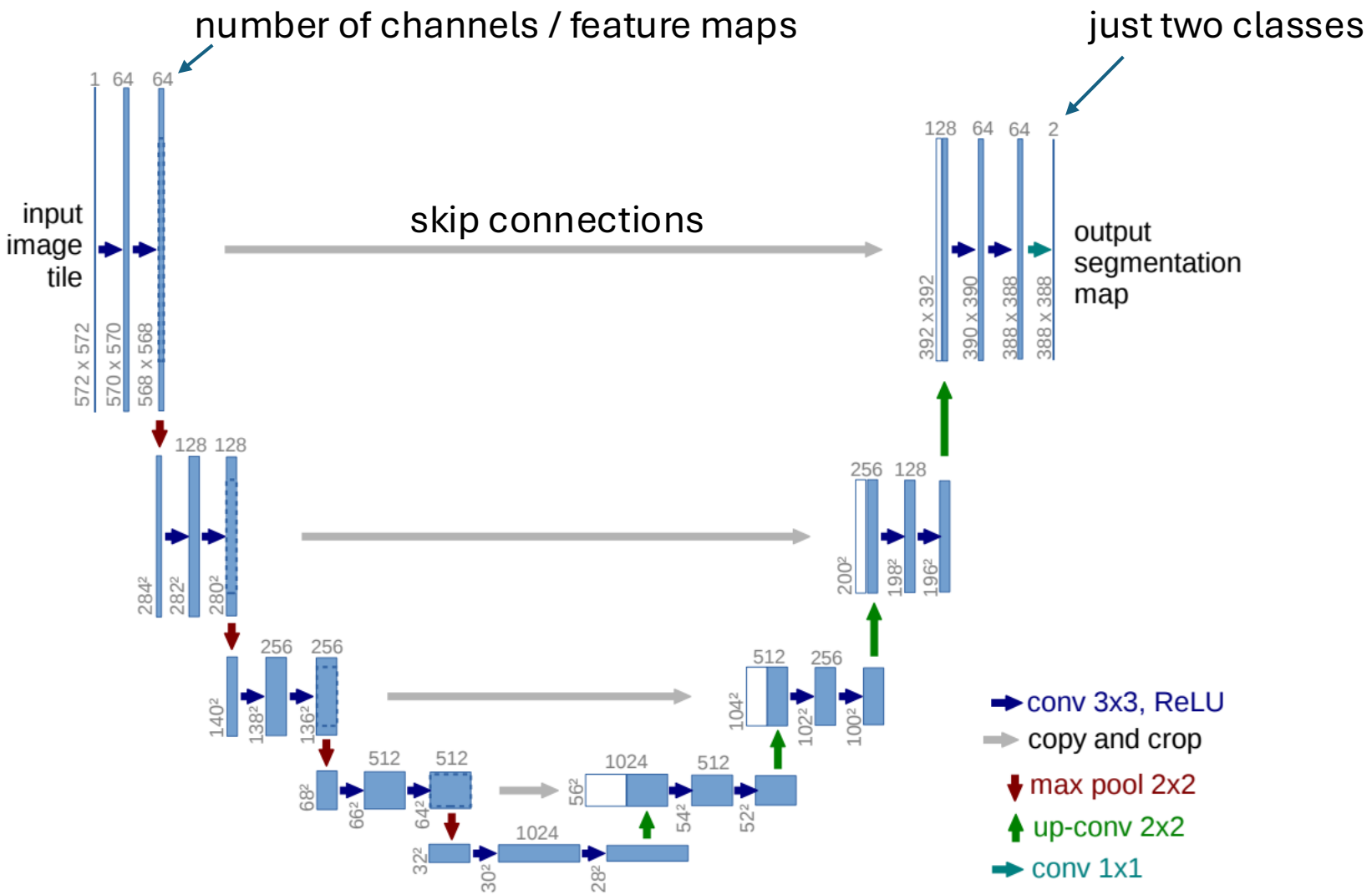
U-Net



also used for learning of depth maps, image synthesis (diffusion), ...

Aside: Autoencoders

Autoencoder

(deep) encoder network

(deep) decoder network

learned together by minimizing differences between original input and reconstructed input (expressed as losses)

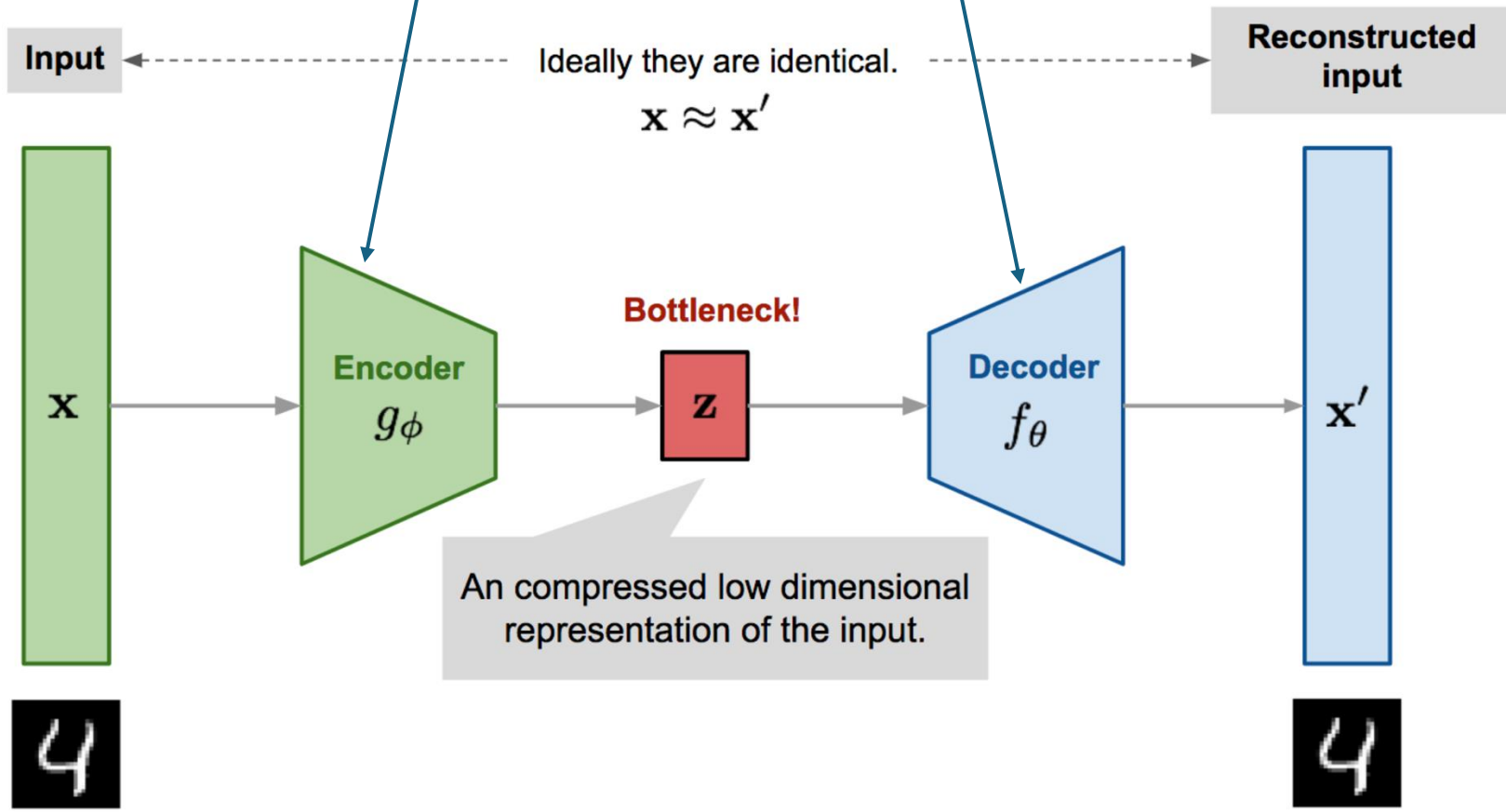
compressed intermediate representation: dimensionality reduction

(alternative to PCA)

for images:

convolutions

transposed convolutions



unsupervised learning

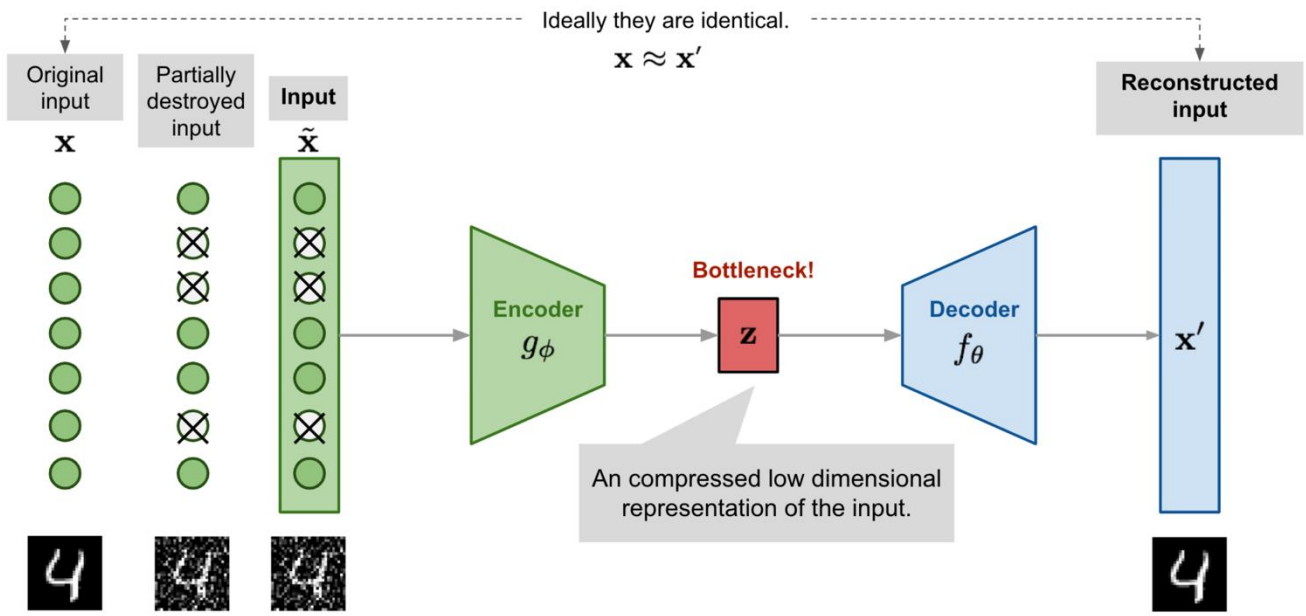
source

Denoising Autoencoder

goal: avoid overfitting and improve robustness of plain autoencoder

learn to remove noise of distorted input $\tilde{x}$ → restore original input x

similar to dropout



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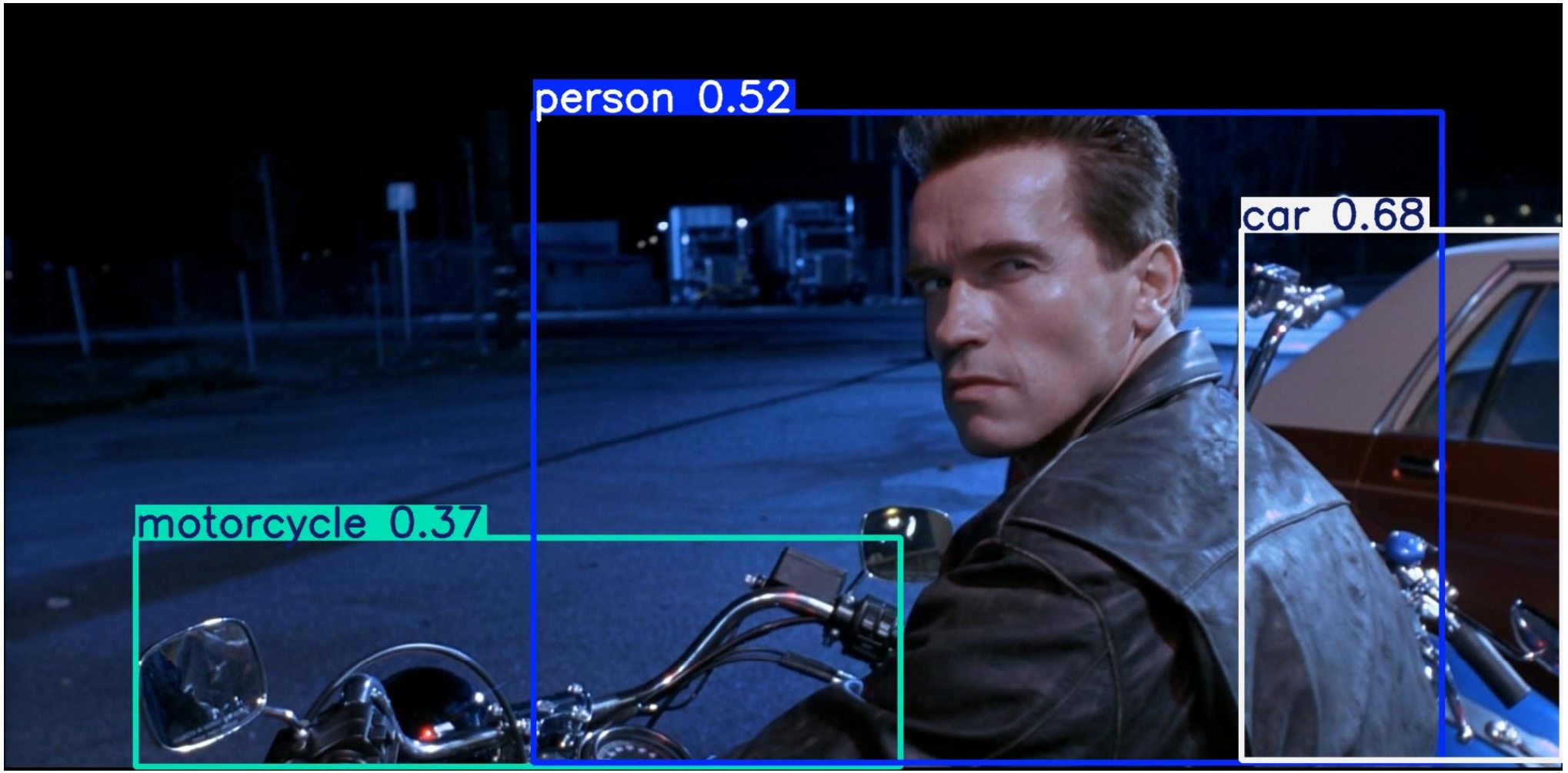
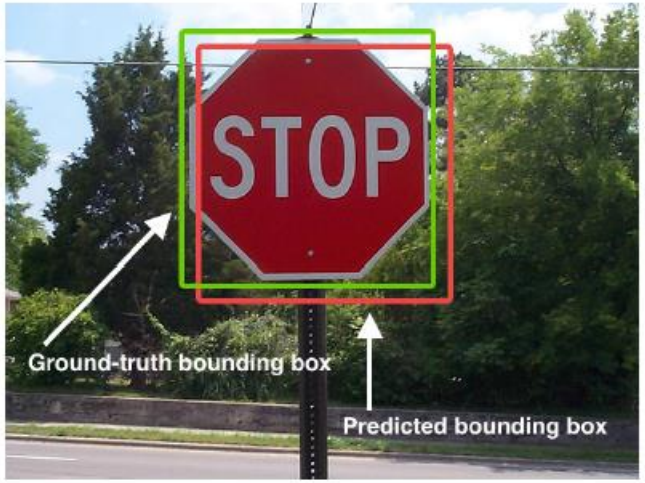
alternative to deconvolution (image restoration)

Object Detection

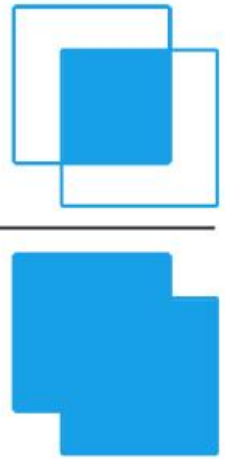
output: bounding boxes with category label and confidence



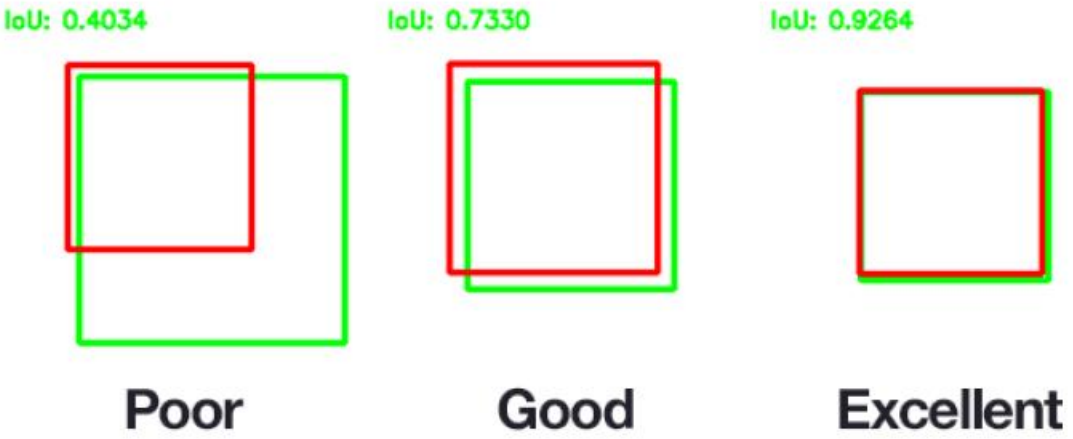
Performance Evaluation of Localization



$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$



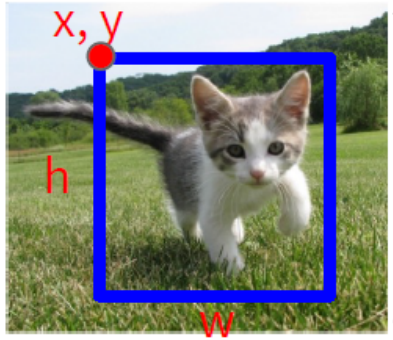
Intersection-over-Union

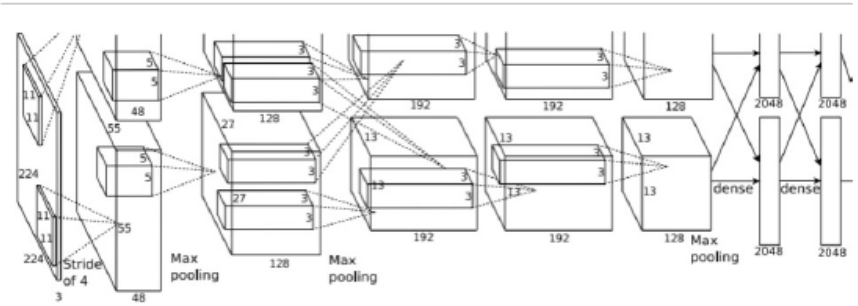


images with multiple objects: number of detections dependent on used confidence threshold

Object Detection: Single Object

(Classification + Localization)





Vector: 4096

Fully Connected: 4096 to 1000

Class Scores
Cat: 0.9
Dog: 0.05
Car: 0.01
...

Softmax Loss

Correct label: Cat

Fully Connected: 4096 to 4

Box Coordinates (x, y, w, h)

L2 Loss

Correct box: (x', y', w', h')

Multitask Loss

+ → Loss

Treat localization as a regression problem!

too complicated for multiple objects

Localization for Multiple-Object Images

idea: classify many different crops of the image as object or background

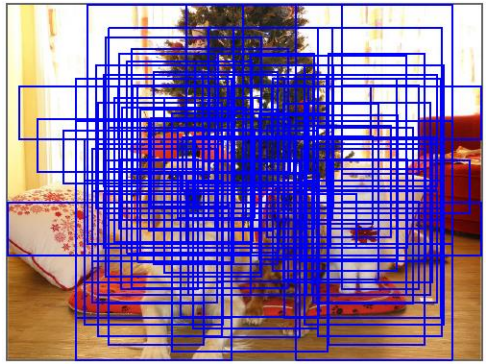
(crops: sliding window over image, iterated at multiple window sizes)

Dog? No
Cat? No
Background? Yes

Dog? Yes
Cat? No
Background? No

Dog? Yes
Cat? No
Background? No

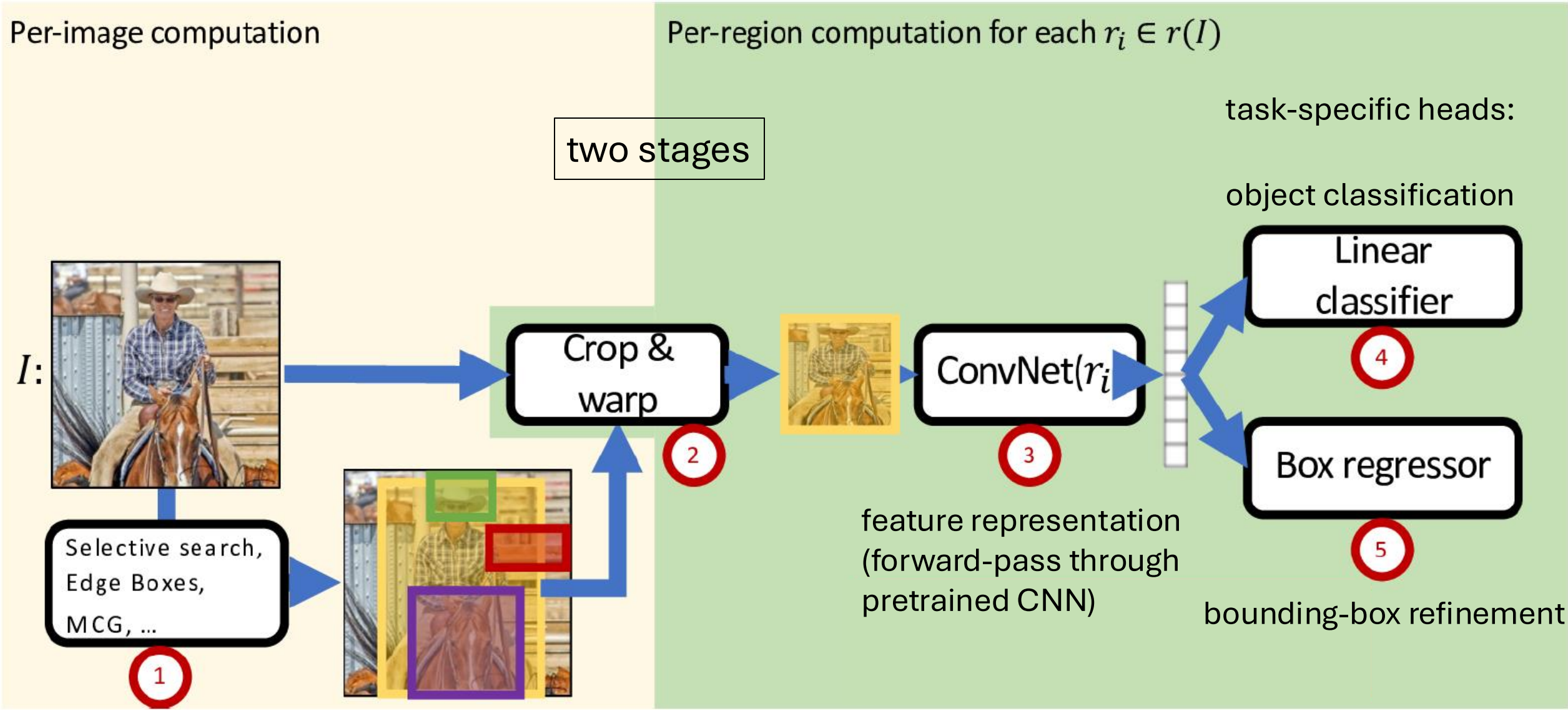
Dog? No
Cat? Yes
Background? No



issue: too many possible crops

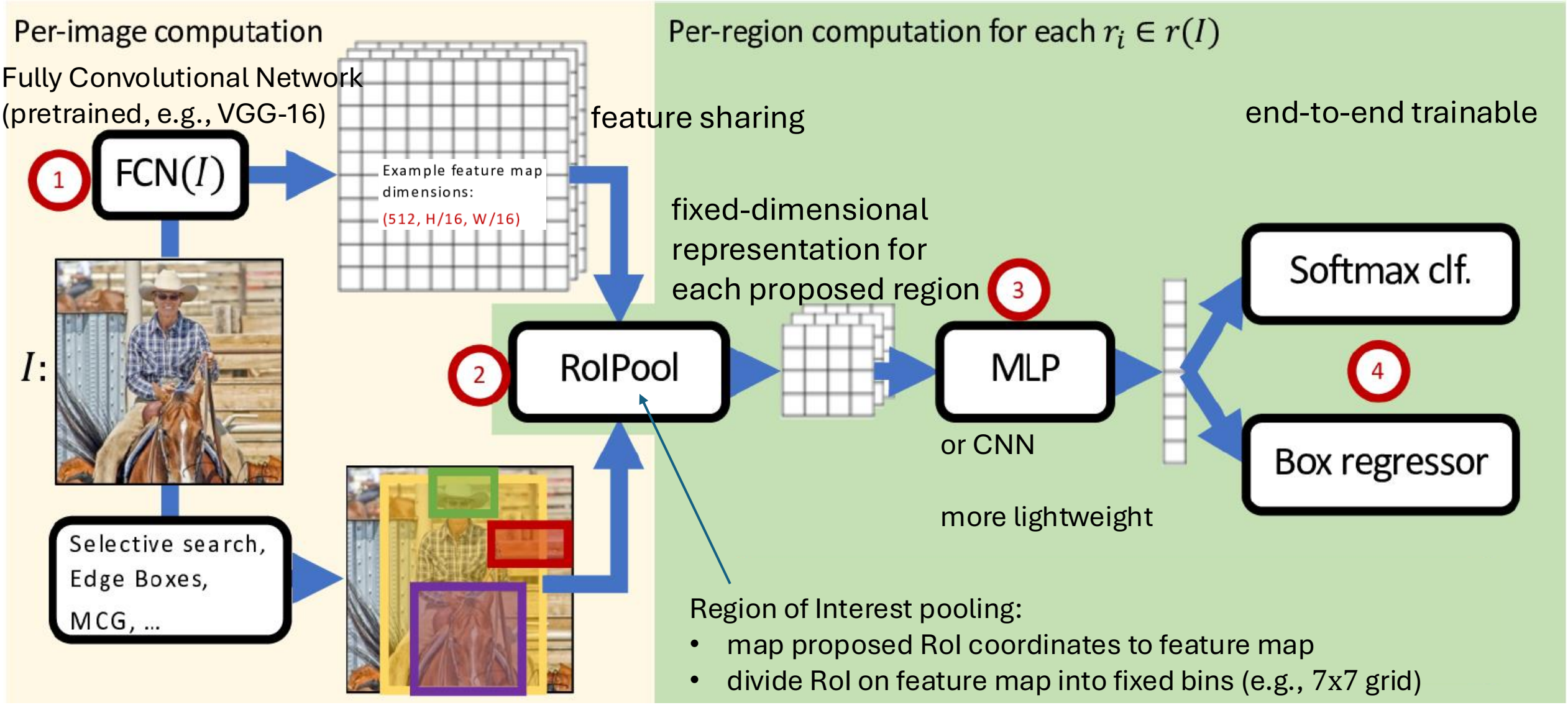
→ need for region proposals

Region-Based Convolutional Neural Network (R-CNN)

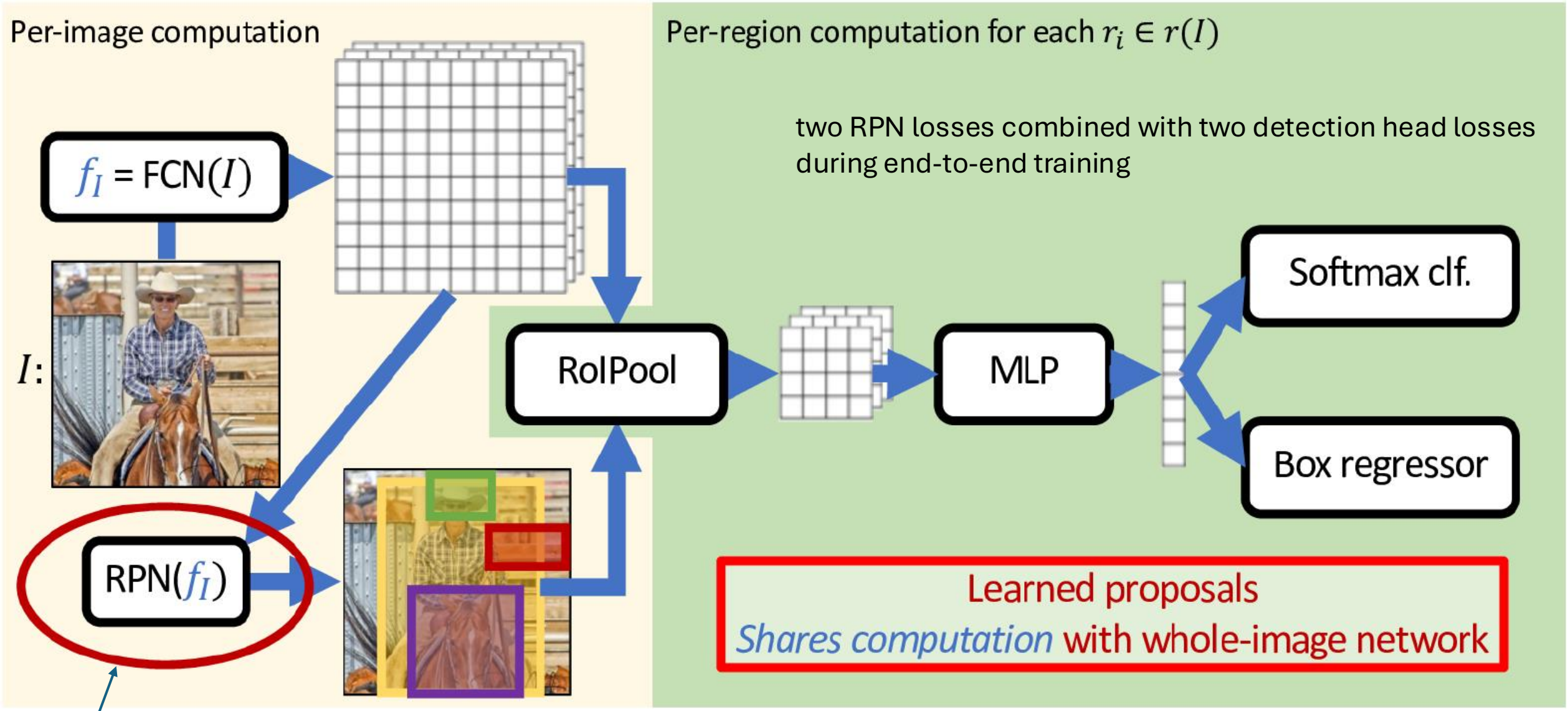


region proposals by "classic" method (~2k)

Fast R-CNN: Crop ConvNet Features



Faster R-CNN: Use Region Proposal Network



binary classification (object or not) with sliding window (e.g., 3x3) over feature map, refining predefined anchor boxes

Feature Pyramid Network (FPN)

different intermediate layers of same backbone CNN

Per-image computation

Per-region computation for each $r_i \in r(I)$

high-resolution features improve detection of small objects

$f_I = \text{FPN}(I)$

I:

RPN(f_I)

RoIPool

MLP

Softmax clf.

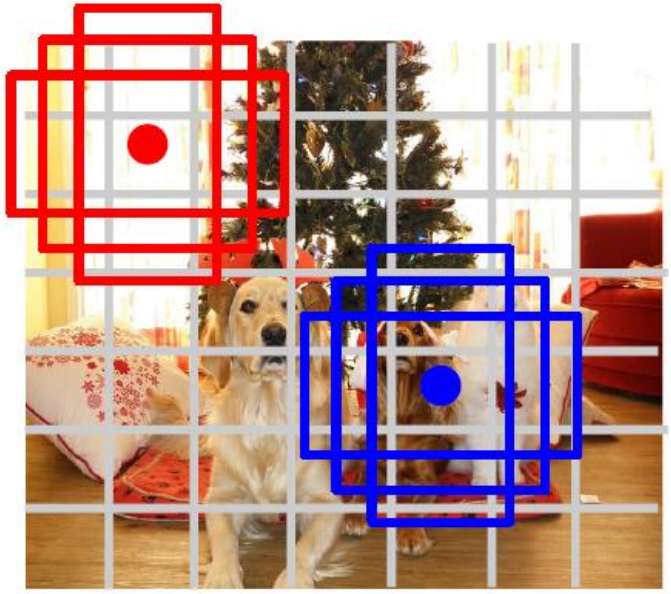
Box regressor

The whole-image feature representation can be improved by making it *multi-scale*

using different anchor sizes for different pyramid levels

Single-Stage Detectors: Drop Per-Region Computation

Input image
3 x H x W



Divide image into grid
7 x 7

or more

Image a set of base boxes
centered at each grid cell
Here B = 3

Within each grid cell:

- Regress from each of the B base boxes to a final box with 5 numbers:
  (dx, dy, dh, dw, confidence)
- Predict scores for each of C classes (including background as a class)
- Looks a lot like RPN, but category-specific!

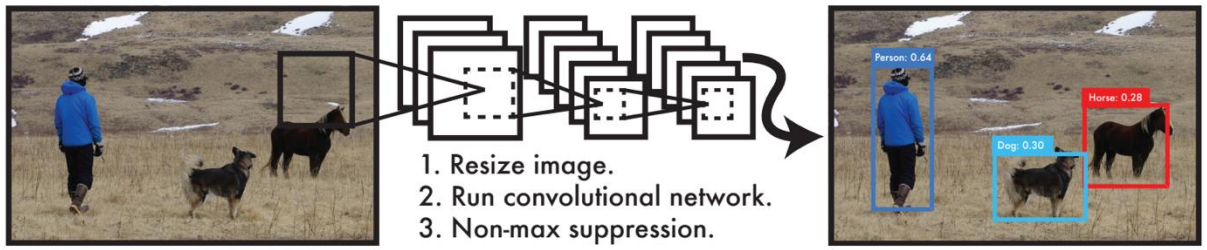
Output:
7 x 7 x (5 * B + C)

Redmon et al, "You Only Look Once:
Unified, Real-Time Object Detection", CVPR 2016
Liu et al, "SSD: Single-Shot MultiBox Detector", ECCV 2016
Lin et al, "Focal Loss for Dense Object Detection", ICCV 2017

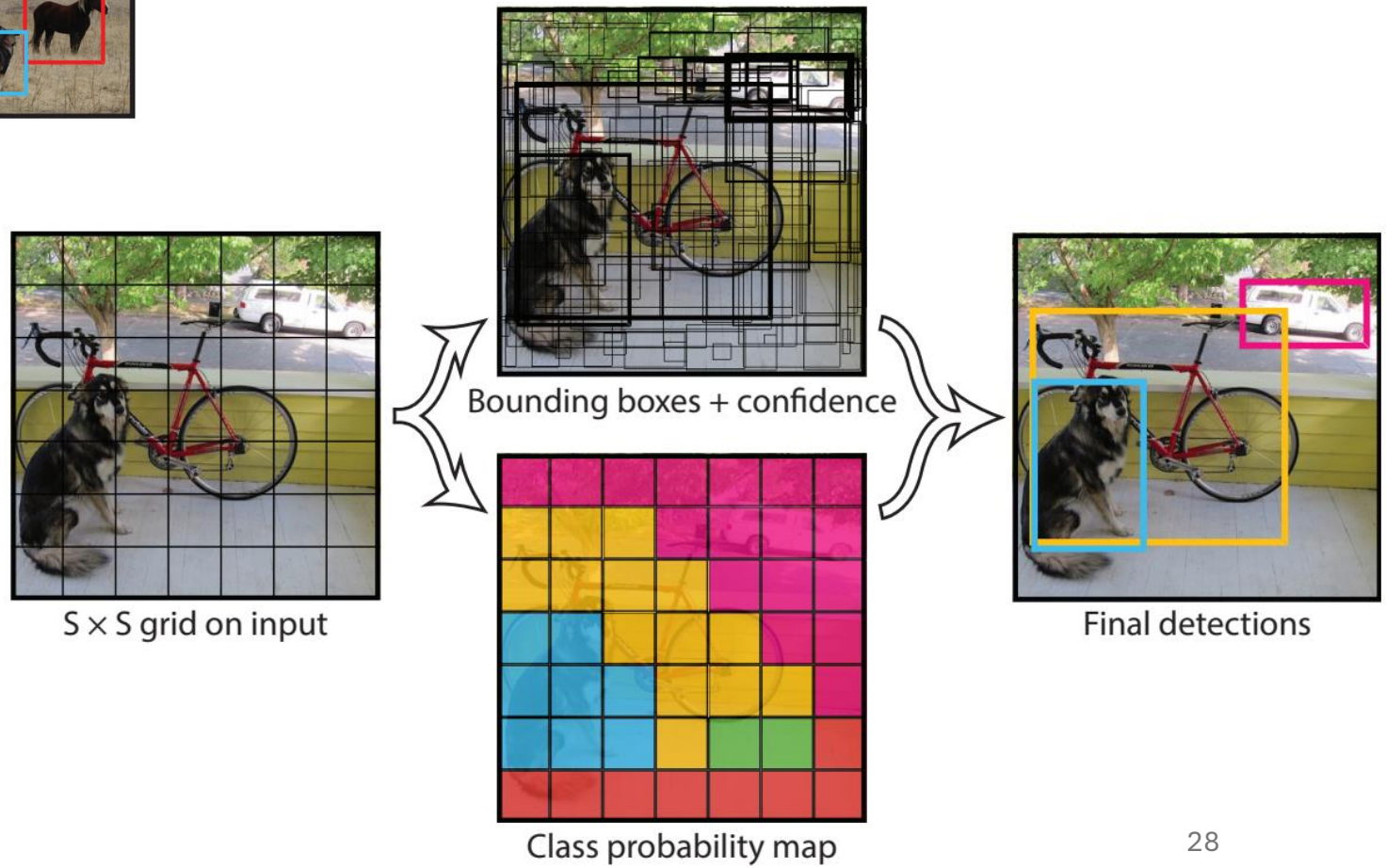
faster, but usually worse performance

YOLO: Real-Time Object Detection

anchor-free approach
in newer YOLO versions



You Only Look Once:
prediction of bounding boxes
and class probabilities in one go



Object Tracking

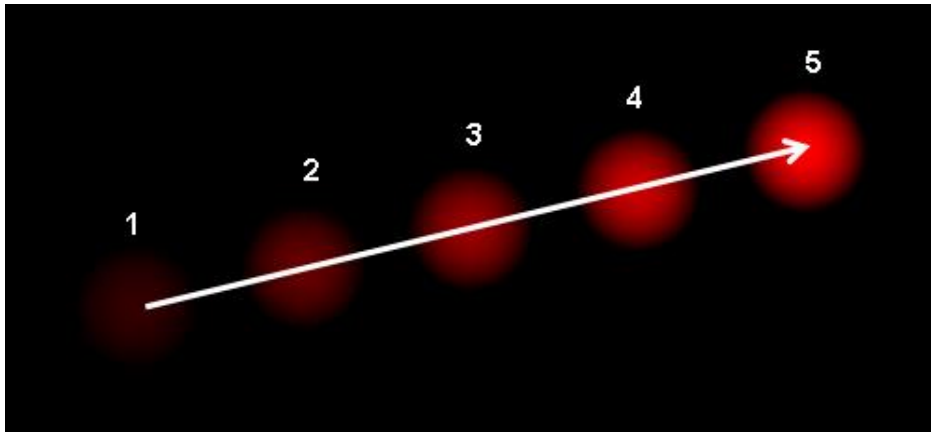
task: locating an object in successive video frames

basic approach:

1. detection: localize target object(s) (e.g., YOLO)
2. motion estimation: predict next location (e.g., Kalman filter or optical flow)
3. appearance matching: comparison of previous and predicted next location (e.g., feature descriptor, CNN features)

detection repeated frequently to correct for accumulated deviations

Optical Flow



dense optical flow:

estimate motion vector of every pixel

e.g., Horn-Schunck method or Lucas-Kanade (LK) method

sparse optical flow:

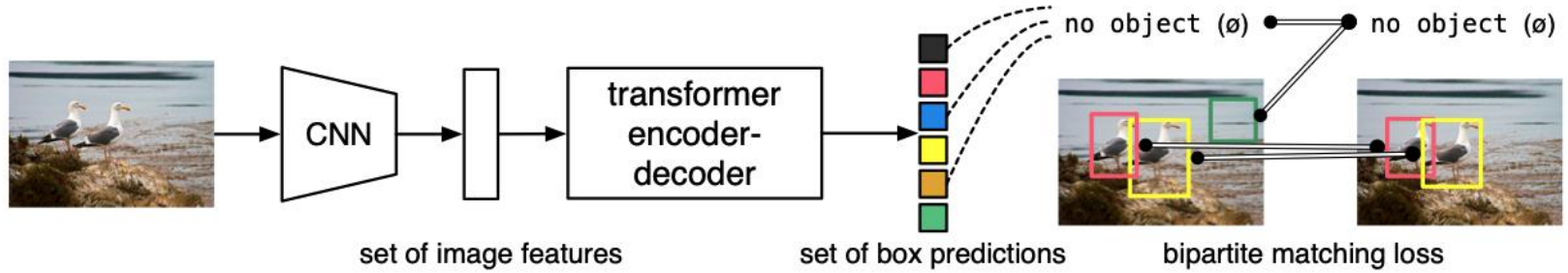
estimate motion vector of few image features

e.g., Kanade-Lucas-Tomasi (KLT) feature tracker

Detection with Transformers

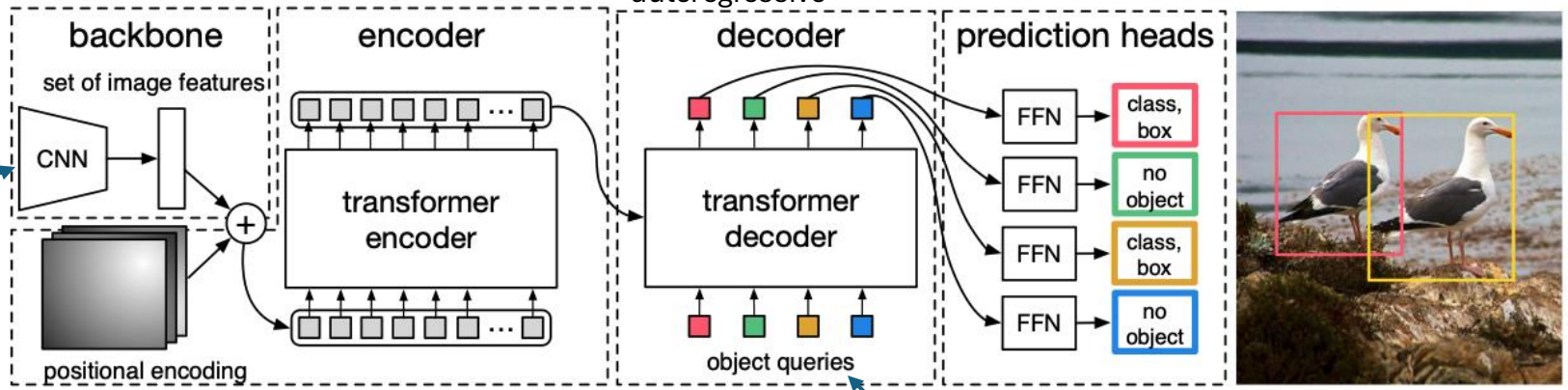
DETR:

idea: use a transformer to directly predict object bounding boxes and class labels from image features (no RPN)



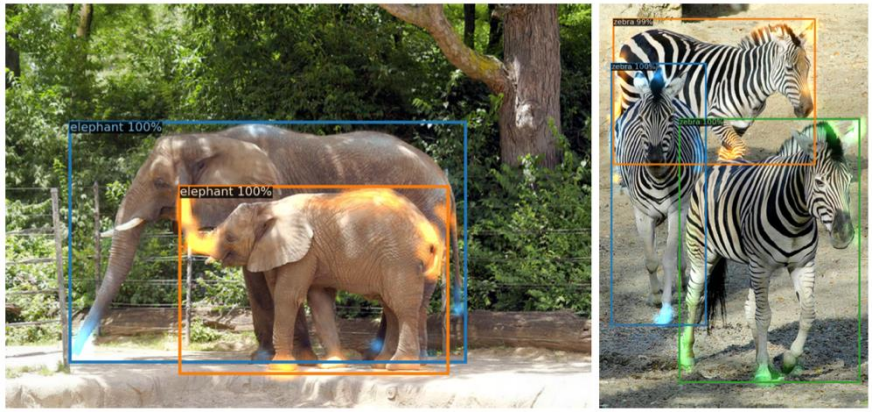
parallel rather than autoregressive

backbone can also be vision transformer



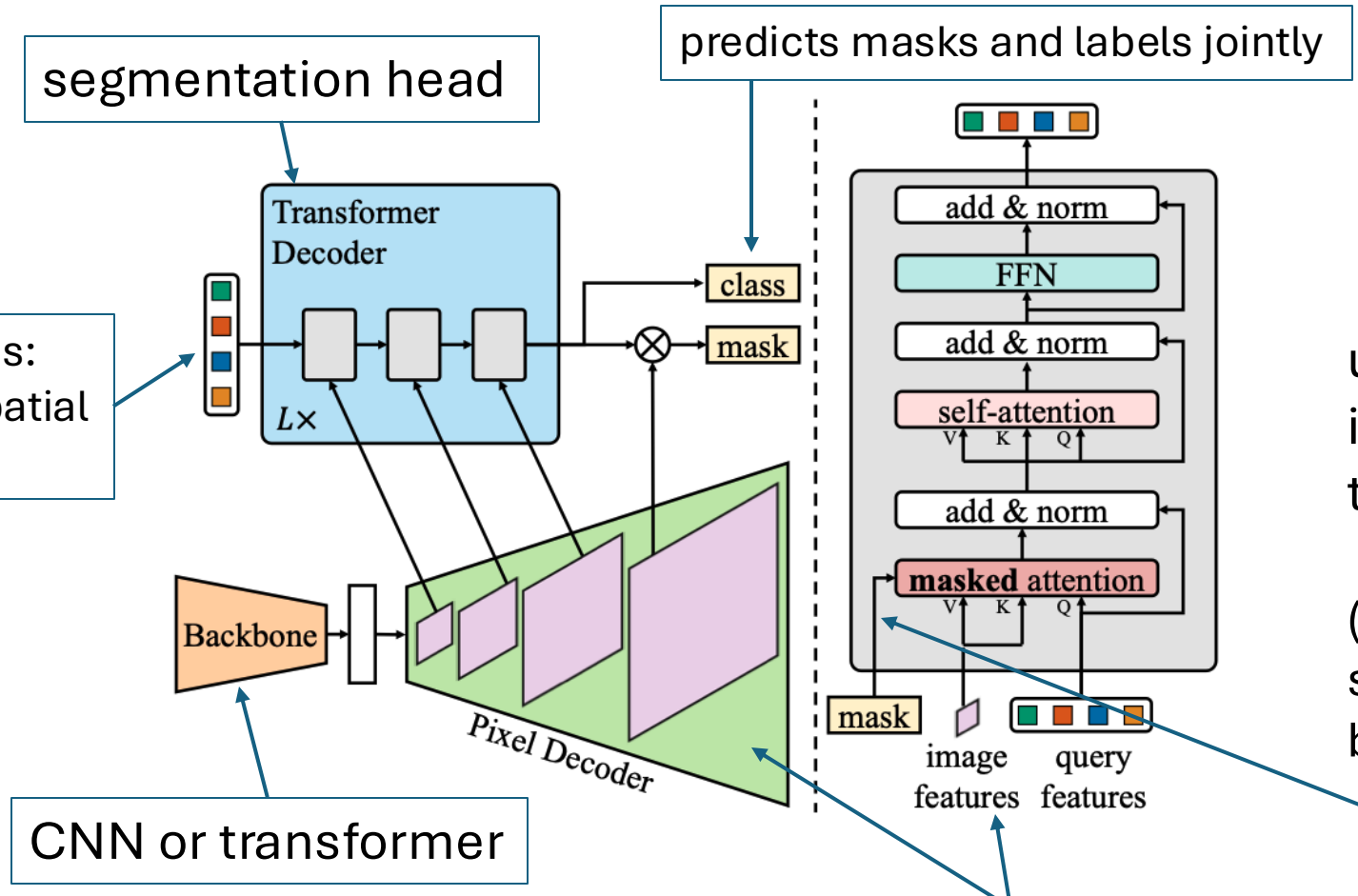
→ popular combination: DINOv2 + DETR

visualized decoder attention for predicted objects:



slot for learnable vectors (“Is there an object here?”)

Segmentation with Transformers



Mask2Former

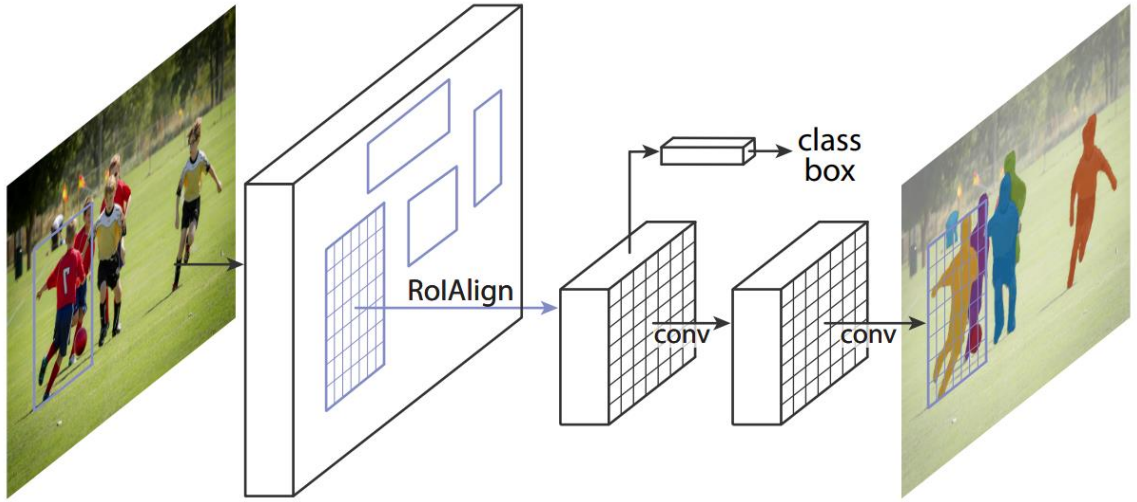
unlike pixel-wise classification, it predicts a set of masks, each tied to an object or region

(similar to DETR, but predicting segmentation masks instead of bounding boxes)

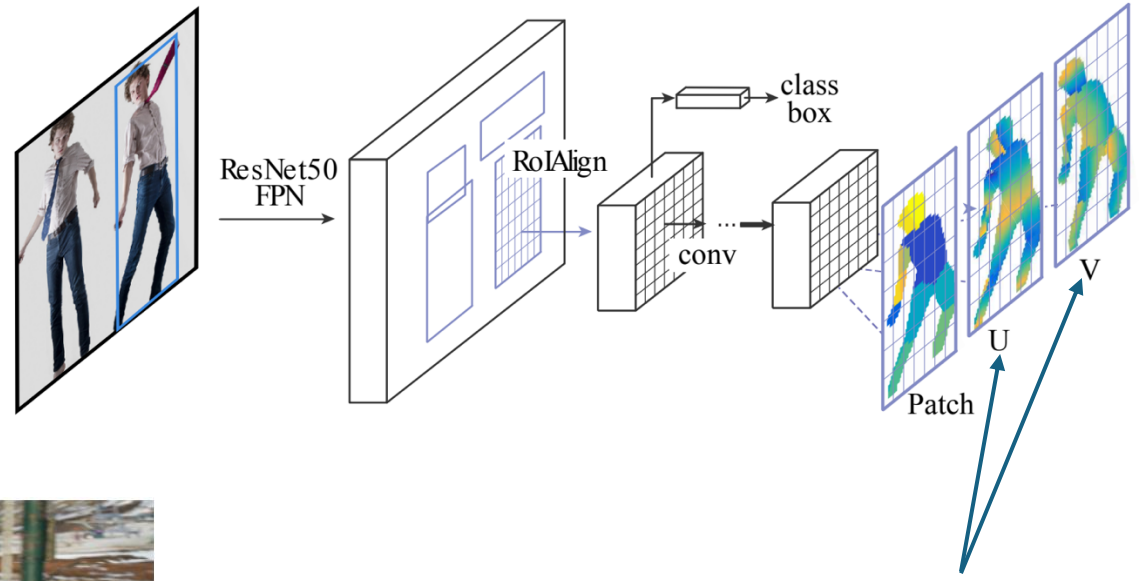
Instance Segmentation

Add Additional Network Heads to R-CNN

instance segmentation (Mask R-CNN):



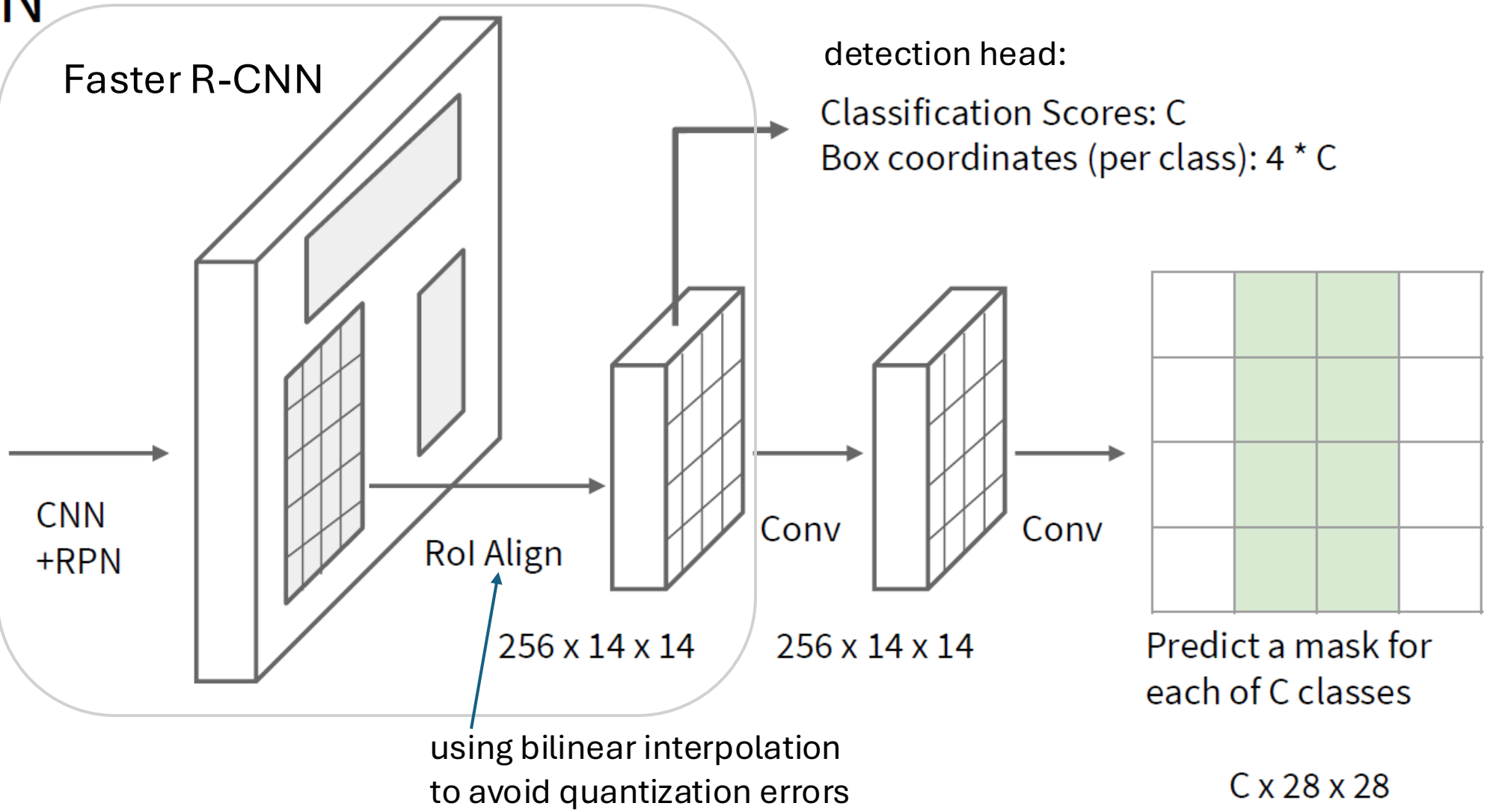
pose estimation (DensePose):



3D model's surface to 2D

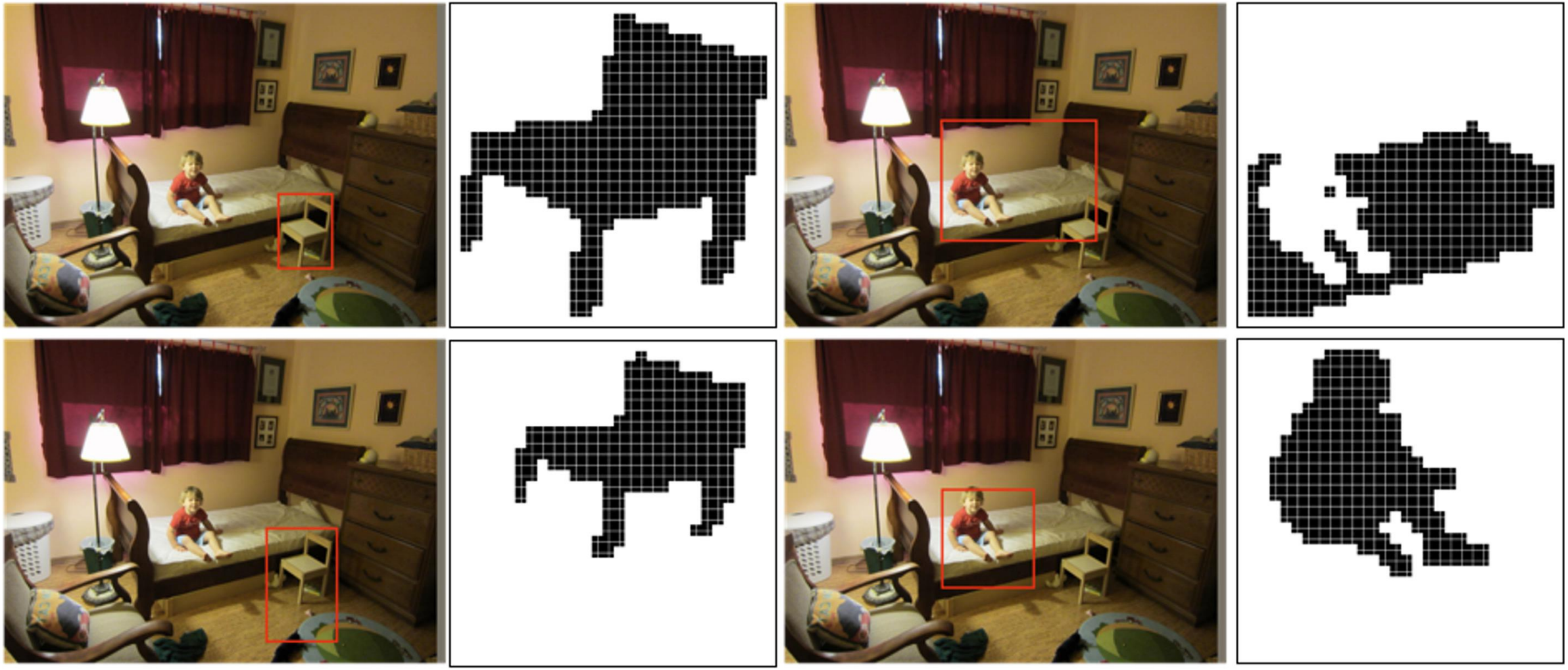
→ total loss is sum of RPN, detection-head, and mask-head losses

Mask R-CNN

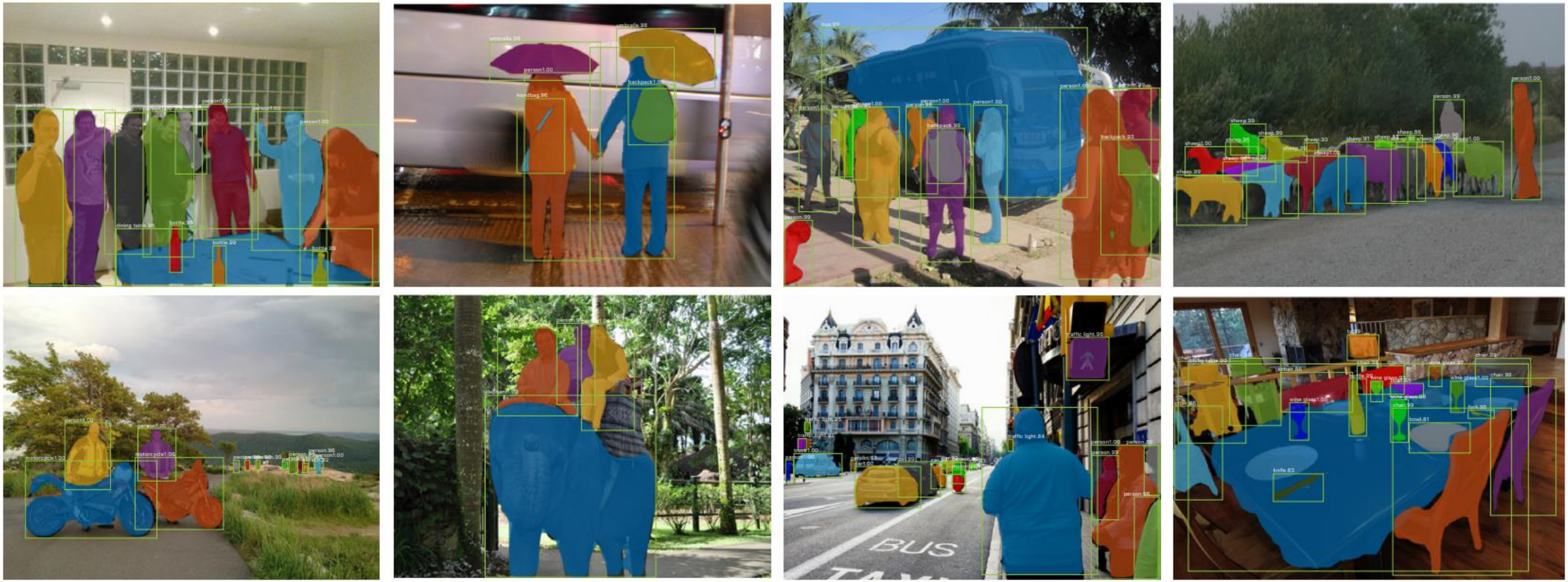


1. object detection (boundary boxes)
2. prediction of separate binary masks for each detected object (specific instances)

image with training proposal | 28 × 28 mask target | image with training proposal | 28 × 28 mask target



results on MS COCO data set



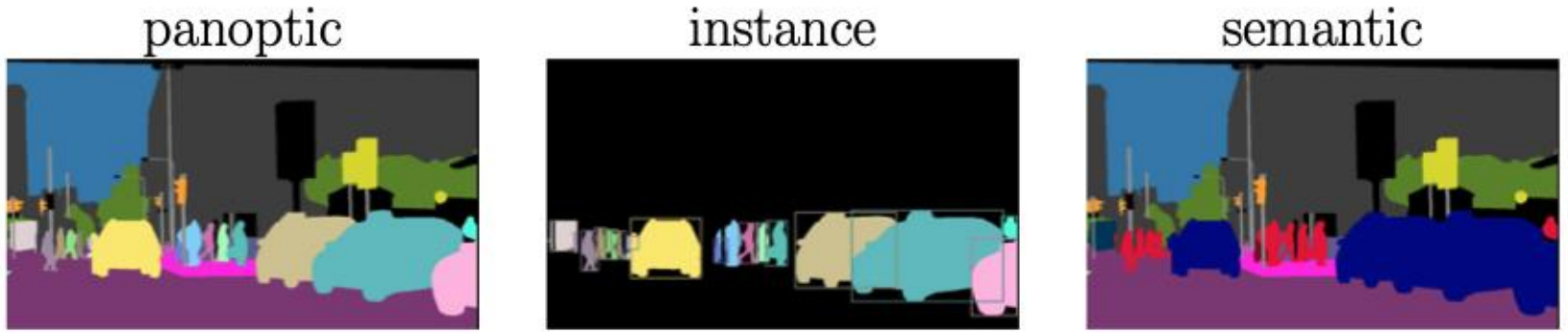
Panoptic Segmentation

unification of semantic sementation (labeling all pixels) and instance segmentation (identifying individual objects)

each pixel is assigned

- a class label (e.g., road, car)
- an instance ID, if it belongs to a “thing” class (e.g., car #1, car #2)

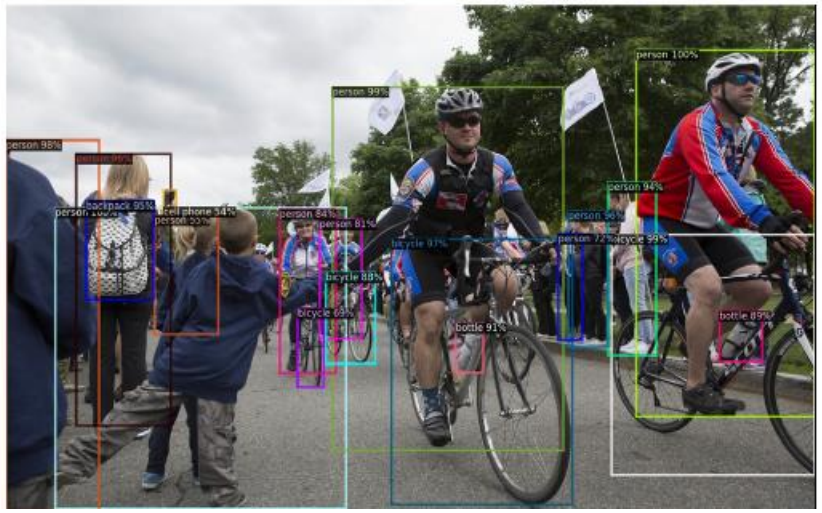
covering both “thing” classes (countable objects) and “stuff” classes (uncountable regions like sky, grass, road)



source

e.g., with combination of Mask R-CNN and fully-convolutional head (or DETR variants such as Mask2Former)

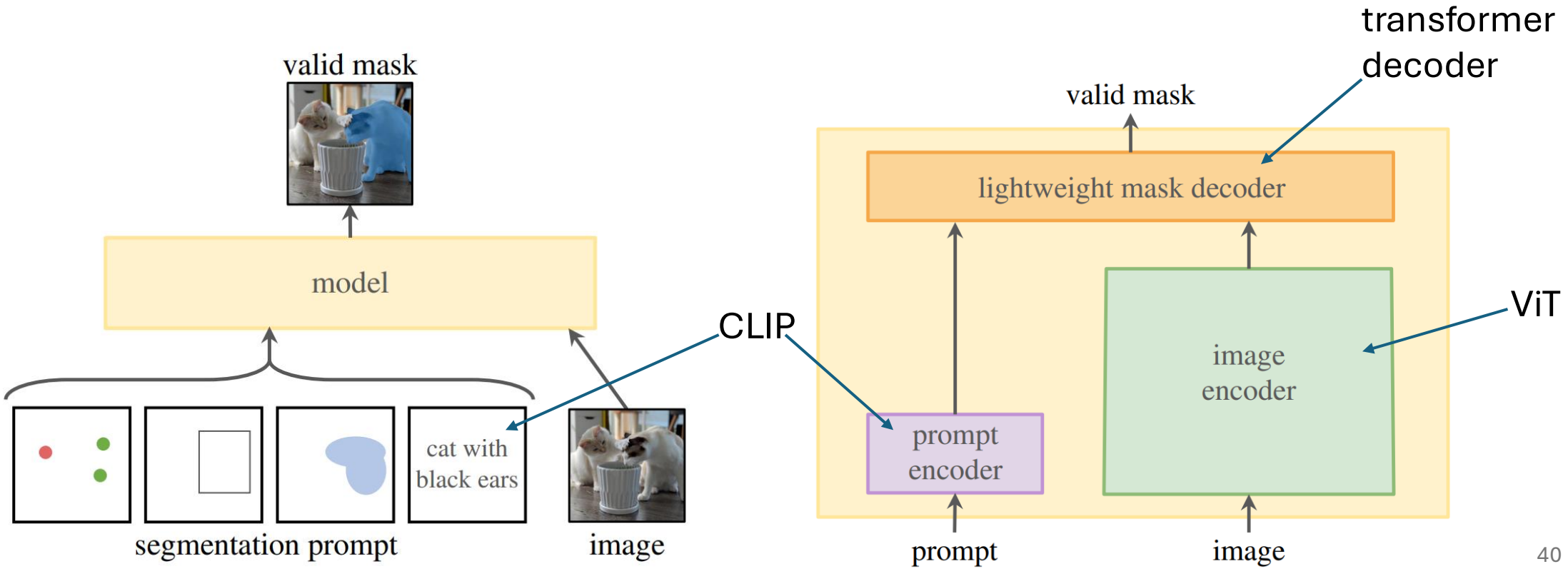
Detectron2: PyTorch Implementations



Promptable Segmentation with Transformers

Segment Anything Model (SAM)

SAM2 includes also video segmentation



Some Off-the-Shelf ML Methods

multivariate tabular data prediction: HistGradientBoosting from scikit-learn (regression, classification)

image classification: feature extraction from DINOv2 or finetune a ResNet/ViT from torchvision (zero-shot: CLIP)

object detection: Faster R-CNN ResNet-50 FPN from torchvision (zero-shot: YOLOv8)

semantic segmentation: FCN ResNet-50 from torchvision (zero-shot: YOLOv8-seg)

instance segmentation: Mask R-CNN ResNet-50 FPN from torchvision (zero-shot: YOLOv8-seg or SAM)

text classification: finetune DistilBERT from transformers (zero-shot: prompt an Ollama LLM)